Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: the kernel disappears under the RES evaluations. Both the bounded and the geometric Euler formulas annihilate the entire unmeasured tensor kernel. Consequently, this failure of tensor injectivity creates no further ambiguity in the retraction to RES.

Keep the original ground fixed and put

$$T = \mathbf{K}\,\mathrm{RES}[*] \otimes_{\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}\,\Gamma[*], \qquad \overline{R} = !\mathbf{K}\,\mathrm{RES}[*], \qquad \tau = \iota([\mathbb{G}_m]_1).$$

The existing quotient by $!I[*]$ is retained. For either $\chi = \chi_g$ or $\chi = \chi_b$, define on homogeneous pure tensors

$$\widetilde{\epsilon}_\chi([X]_r \otimes [U]_m) = \iota([X]_r)\tau^m \chi(U) \in \overline{R}_{r+m}.$$

The finite-Gamma balancing is valid: a finite degree-l Gamma object with N points has Euler characteristic N, while its RV lift has class $N\tau^l$ in $\overline{R}$. This produces a graded ring homomorphism on T independently of any claim that Ψ is injective.

For every generator of the kernel found above,

$$g = [X]_r \otimes \big([U]_m - [U + \delta e_i]_m\big), \qquad c : X \longrightarrow \delta^\sharp \text{ definable},$$

one has

$$\widetilde{\epsilon}_\chi(g) = \iota([X]_r)\tau^m\big(\chi(U) - \chi(U + \delta e_i)\big) = 0.$$

Both Euler characteristics are invariant under translations of semilinear Gamma sets; the translation need not have an RV lift over the original ground. Thus

$$\boxed{\ker \Psi \subseteq \ker \widetilde{\epsilon}_b \cap \ker \widetilde{\epsilon}_g.}$$

This is direct vanishing on generators, already before localization. No torus class has been cancelled.

The construction therefore descends. Surjectivity gives unique homomorphisms

$$\epsilon_\chi : \mathbf{K}\,\mathrm{RV}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*], \qquad \epsilon_\chi \circ \Psi = \widetilde{\epsilon}_\chi.$$

Choose any tensor preimage of an RV class and apply the Euler formula; the displayed kernel inclusion makes the result independent of that choice. In the construction in § 4.1, the use of Ψ^{-1} can therefore be replaced by descent through the surjective Ψ. No additional quotient of RES is required beyond the existing $!I$ relations needed for balancing.

Writing $s = [1]_1$ and $\mathbb{L}_1 = \tau + s$, the subsequent normalizations multiply a degree-n value by s^{-n} for $\mathbb{E}_b$ or by $\mathbb{L}_1^{-n}$ for $\mathbb{E}_g$. Hence both normalized maps still kill the tensor kernel. For the fifth-root example, the individual terms $[R]_0 \otimes [I]_1$ and $[R]_0 \otimes [J]_1$ both have bounded Euler image zero and geometric Euler image $-\iota([R]_0)\tau$ before normalization. Their nonzero difference is invisible to both.

This preserves the Euler-retraction construction against the T001 obstruction; it does not certify other statements in § 4. The kernel and descent calculations remain mathematical proofs rather than completed Lean exports. The complete argument is recorded in the section's retraction verification note. No HK comparison theorem has been taken as a premise, and the approved working manuscript is unchanged.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of $\mathrm{RV}[*]$ then it may also denote the object $(\boldsymbol{U}, 0)$ of $\mu\mathrm{RV}[*]$ with the constant volume form 0.

Notation 3.20. Let $[1] \in \mathbf{K}_+\,\mathrm{RES}[1]$ be the class of the singleton $\{1\}$. The class of the singleton $\{1\}$ in $\mathbf{K}_+\,\mathrm{RES}[0]$ is the multiplicative identity of $\mathbf{K}_+\,\mathrm{RES}[*]$ and hence is simply denoted by 1. We have $[\mathrm{RV}^{\circ\circ}_\infty] = [\mathrm{RV}^{\circ\circ}] + 1$ in $\mathbf{K}_+\,\mathrm{RV}[\leq 1]$.

Let I_{sp} be the (nonhomogeneous) semiring congruence relation on $\mathbf{K}_+\,\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}_\infty])$. Let

$$\boldsymbol{P} = [1] - [\mathrm{RV}^{\circ\circ}] \in \mathbf{K}\,\mathrm{RV}[1].$$